Problems Involving Binary (or maybe other bases)

1. Find all sets of nonnegative integers S satisfying: for any nonnegative integer a, there exists a unique ordered pair (x,y) chosen from S (x and y can be equal) such that $a=x+2y$.

2. Is it possible to partition the set of nonnegative integers into two sets S and T, such that for any positive integer n, .the amount of ways to express n as two **distinct** numbers in S is equal to the amount of ways to express n as two **distinct** numbers in T?

3. There are 5 balls, each of them can be either 100 grams or 95 grams, and it is known that the majority is 100 grams. You have an electronic scale that can weigh one or more balls, but the display is broken and can only show the units digit. Find the least amount of weighings that can determine the masses of all balls. If you are required to use a "fixed" theme (first name the sets of balls to weigh each time, then weigh the sets one by one), how many themes are there?

4. There are $\frac{3^n-3}{2}$ balls, among which all but one of the balls have the same mass. The special ball can be either heavier or lighter than the others. With a balance, find the special ball in n weighings, and determine whether it is heavier or lighter than the others. Also, you can use a "fixed" scheme of weighing, which means you can predetermine which balls are weighed at which time all at once, without needing to pick balls depending on previous results.

5. A row of n coins are numbered 1 through n. It is known that there are exactly two heavy coins, which are 11 grams each, and are **adjacent**. Normal coins are 10 grams each. You have an electronic scale that can weigh the total weight of any amount of coins. If you are allowed to weigh at most m times, and you must determine which coins to weigh each time beforehand (i.e. you cannot pick coins that depend on the result of previous results). Find the maximum of n (in terms of m) such that you are guaranteed to find the two heavy coins.

6. Given an integer $n \geq 3$, Find the amount of 5-subsets of $\{1,2,...,2^n-1\}$ whose bitwise XOR equals 0.

7. Find all integers $n \geq 7$ such that their exists a class $\boldsymbol{F}$ of 3-element subsets of $\{1, 2, ..., n\}$ that satisfies the following condition:
For any two different elements a and b in S, their is exactly one set in $\boldsymbol{F}$ containing both a and b.
For any six different elements a, b, c, x, y, z in S, if $\{a,b,x\}$, $\{b,c,y\}$, $\{c,a,z\}$ are in $\boldsymbol{F}$, then $\{x,y,z\}$ is in $\boldsymbol{F}$.

8. Let $S=\{0,1,2,\cdots,2^n-1\}$ where n is an integer larger than 1. Color all the 3-subsets of S so that any two of them with exactly 1 common element have different colors. Find the minimum amount of colors needed.

9. There are 8 multiple-choice problems in a test, each problem has only two options A and B that the participants need to choose from. All participants have made choices in all problems without leaving blank. For any two participants, when looking at their choices in the 8 problems, (A, A), (A, B), (B, A), (B, B) each appear in exactly 2 problems. Find the maximum amount of participants.

10. S is the set of all binary strings of length 31. T is a **maximum subset (note: distinguish from "maximal subset")** of S such that for any two strings in T, their Hamming distance is at least 3. Does T necessarily consist of pairs of complementary strings?

11. Nina and Tadashi play the following game. Initially, a triple (a, b, c) of nonnegative integers with $a+b+c=2021$ is written on a blackboard. Nina and Tadashi then take moves in turn, with Nina first. A player making a move chooses a positive integer k and one of the three entries on the board; then the player increases the chosen entry by k and decreases the other two entries by k. A player loses if, on their turn, some entry on the board becomes negative.
Find the number of initial triples (a, b, c) for which Tadashi has a winning strategy.

12. A round table has 32 seats which are marked with number 1 to 32. On the table there are 32 lamps near each seat. Some lamps are not turned on. The host writes down a list of numbers and asks the servant to flip the switches of the corresponding lamps according to the list. An evil person is able to see the host's list and rotate the table before the servant gets there. So lamps are rotated from their original position but each lamp is still near a seat. Assuming the servant only flips the switches according the list blindly. Prove that after some rounds, eventually the host can turn on all the lamps.

13. (1)There are 5 multiple-choice problems in a test, each problem has four options A, B, C, D that the participants need to choose from. All participants have made choices in all problems without leaving blank. For any two participants, when looking at their choices in the 5 problems, there is exactly one problem that they chose the same option. Find the maximum amount of participants.

(2)Suppose that the amount of participants takes the maximum above. Does there exist a positive integer n such that it is possible to run n editions of such tests so that for any three participants, their is exactly one problem that they chose the same option in all $5n$ problems?

14. There are some kings on a chessboard, all initially placed in different squares. Alnardo and Bernaldo play alternately, with Arnaldo starting. On each move, one of the two players chooses a king and can move it one square to the right, one square up, or one square up to the right. In the event that a king is moved to an occupied square, both kings are removed from the game. The player who can remove two of the last kings or leave one last king in the upper right corner wins the game. Determine the positions such that the second player wins.

15. The numbers 1 through 50 are written on the blackboard. Two players take turns to move. Each player can erase 3 or 4 consecutive numbers. The player who cannot make any moves loses. If the first player uses any of his possible winning strategies, find the minimum of the smallest number he can erase on the first step.

16. On the lattice grid of the plane, walk from the origin along the sides of the cells, each step either goes right by 1 or up by 1, and stop after walking a certain amount of steps. For each such path, draw a "worm" that contains the cells that share at least one **vertex** with the path. Prove that for any integer $n>2$, the amount of worms that can be divided into 1×2 dominoes in exactly n ways is equal to $\varphi(n)$。

17. There are finitely many real numbers on the blackboard. Players Sam and Tom take turns to make moves. In each move, Sam changes one of the numbers to a rational number smaller than it, and Tom changes one of the numbers to a rational number larger than it, with these restrictions:

a)If the number is an integer, the player can only change to an integer with smaller absolute value.

b)If the number is a rational number but not an integer, the player can only change to a rational number with smaller denominator (when written in reduced fractions with positive denominator).

The player who cannot make a move loses. The players toss a coin to determine who moves first. A game is called "balanced" if whoever moves second has a winning strategy.

(1)Find all real numbers x such that the game $(\sqrt{2},x)$ is balanced.

(2)Find all real numbers y such that the game $\left(\frac{3}{2},y,y\right)$ is balanced.

(3)Does there exist a real number z such that the game $(2,z,z,z)$ is balanced?